CHEW SIN AI

SOFTWARE ENGINEER · FULL-STACK & AUTOMATION

Johor Bahru, Johor, Malaysia • +60 17-742 3008 • chewsinai2002@gmail.com

Portfolio: chewsinai02.github.io/portfolio • **GitHub:** github.com/chewsinai02 • **LinkedIn:** linkedin.com/in/sin-ai-c-41810a121

PROFESSIONAL SUMMARY

Results-driven Software Engineer with proven enterprise experience developing high-availability warehouse management systems (WMS), cross-platform mobile scanner applications in Flutter, and robust backend microservices in Java, PHP, and FastAPI. Adept at automating critical logistics operations with cron concurrency locks, building AI-powered OCR document ingestion pipelines, and mentoring university students in computer science fundamentals. Recognized with a Bronze Medal and research publication at InIIC 2024 for developing an award-winning real-time patient care ecosystem.

PROFESSIONAL EXPERIENCE

Software Engineer (Permanent) & Intern | **Tiong Nam Logistics**
Oct 2025 – Present
Johor, Malaysia

- **WMS Handheld Scanner Mobile App:** Architected and deployed a ruggedized Android warehouse scanner application in Flutter with offline-first SQLite caching and hardware barcode triggers, streamlining inbound floor receiving, put-away, outbound picking, and Physical Stock Count (PSC). *[Flutter, SQLite, REST APIs]*
- **Enterprise WMS Web Platform:** Maintained and enhanced core enterprise warehouse management web platform, implementing inventory location transfer modules, diagnosing high-concurrency database locks, and optimizing mission-critical SQL queries. *[Java, PHP, MySQL]*
- **Operations & Billing Automation:** Engineered automated PHP cron background workers executing daily stock transfers, complex logistics billing calculations, and scheduled dispatch notifications with file-lock overlap prevention, eliminating manual data entry delays. *[PHP, MySQL, Linux Cron]*

Part-Time Computer Science Academic Tutor | **Assignment Helper**
2025 – Present
Remote

- **Academic Mentorship:** Mentored university undergraduate students across core CS subjects including Java, Python, Object-Oriented Programming (OOP), Data Structures & Algorithms, and database design.
- **Code Quality & Architecture:** Conducted code reviews, guided debugging sessions, and reinforced software engineering best practices, design patterns (MVC), and clean architecture principles.

Software Support Engineer (Part-Time) | **Rockbell Software Company**
Jun 2025 – Jul 2025
Johor, Malaysia

- **ERP Diagnostics:** Provided frontline technical support and diagnostics for commercial enterprise ERP and accounting software.
- **Database Troubleshooting:** Executed diagnostic SQL queries against client relational databases to troubleshoot data integrity discrepancies and resolve customer bug reports. *[SQL, Relational DBs]*

Professor Assistant · Parallel Computing | **NVIDIA CUDA Program**
2023
Academic

- **GPU Parallel Programming:** Assisted faculty in guiding students through GPU parallel computing architectures, multi-threaded execution models, and CUDA C/C++ kernel acceleration. *[CUDA C/C++, Parallel Computing]*

Part-Time Educator | **Smart Little Beans Early Reading Centre**
2021 – 2022

- Facilitated foundational literacy and structured early learning modules, fostering disciplined communication and patience.

KEY ENGINEERING PROJECTS

Patient Care Real-Time Management System | **Academic Flagship · Bronze Award Winner** **2024**

- Sole architect and developer of an end-to-end hospital communication ecosystem consisting of a 4-role Laravel web portal (Doctor, Nurse, Admin, Family) and a Flutter cross-platform mobile patient application with live Firebase Realtime Database sync, eliminating hospital ward communication bottlenecks. Awarded Bronze at InIIC 2024 with published research thesis. *[Laravel, Flutter, Firebase, MySQL]*

AI Mail & Document Ingestion Pipeline | **Enterprise Logistics [NDA Protected]** **2025 – 2026**

- Constructed an automated document ingestion pipeline that polls incoming operational shipment emails, processes raw PDF/image attachments using Google Gemini multimodal AI for structured OCR key extraction, and uses Playwright headless browser scripts to auto-submit payloads into enterprise compliance portals, saving dozens of manual administrative hours weekly. *[React, PHP, Gemini AI, Playwright]*

AI PDF Split, Merge & Auto-Renaming Engine | **Enterprise Automation [NDA Protected]** **2025 – 2026**

- Developed a high-throughput document intelligence service featuring a FastAPI asynchronous backend and React frontend. Leveraged PyMuPDF and Gemini AI to automatically segment multi-page bulk invoices and delivery orders, parse shipment identifiers, and execute rule-based auto-renaming and folder distribution. *[FastAPI, React, Gemini AI, PyMuPDF]*

Logistics Billing & Scheduled Dispatch Automation | **Enterprise Logistics [NDA Protected]** **2025 – 2026**

- Built a configurable logistics billing calculation engine with custom formula evaluation, rate card table mapping, and automated monthly invoicing summaries. Paired with a scheduled email console utilizing strict OS file-lock concurrency guards to prevent duplicate customer notices. *[PHP, MySQL, Linux Cron]*

API Observability & JSON Log Platform | **Open Source Developer Tool** **2025**

- Created an API observability platform to log, inspect, and diagnose enterprise REST payloads. Flags schema mismatches, missing attributes, and HTTP error codes with real-time UI filtering, enabling operations staff to replay failed JSON payloads without code changes. *[PHP, MySQL, JavaScript]*

TwoGether · Couple Wellness & Nutrition Tracker | **Mobile Application** **2024**

- Built a dual-user shared nutrition and meal tracking mobile app in Flutter, featuring synchronized calorie balance calculations, weight milestone progress charts, and instant push notifications via Firebase Cloud Messaging (FCM). *[Flutter, PHP Backend, Firebase FCM]*

Pet Adoption Management System | **Open Source Desktop Application** **2023**

- Developed a modular desktop management system in JavaFX following the MVC design pattern, featuring role-based authentication, animal medical history records, adoption matching, and relational SQL persistence. *[JavaFX, SQL, MVC Architecture]*

TECHNICAL SKILLS

Programming Languages: PHP, Java, Dart, Python, JavaScript (ES6+), C++, HTML5, CSS3, SQL

Frameworks & Libraries: Flutter, Laravel, FastAPI, React.js, JavaFX, PyMuPDF, Bootstrap, Tailwind CSS

Databases & Cloud: MySQL, SQLite, Firebase (Realtime Database, Firestore, Auth, FCM), RESTful APIs

Developer Tools & AI: Google Gemini AI, Playwright Automation, Linux / Cron Jobs, Git & GitHub, Android Studio, VS Code

EDUCATION

Bachelor of Software Engineering (Honours) | **Southern University College**
2022 – 2025
Johor, Malaysia

- Academic Performance: CGPA 3.70 / 4.00 · Dean's List Award Recipient (Multiple Trimesters)
- Core Coursework: Data Structures & Algorithms, Object-Oriented Software Design, Database Systems, Web Development, Mobile Application Development, Operating Systems

STPM (Sijil Tinggi Persekolahan Malaysia) | **SMK Taman Kluang Barat**
2021 – 2022
Kluang, Malaysia

- Pre-University / Malaysian Higher School Certificate · CGPA: 3.17 / 4.00

SPM (Sijil Pelajaran Malaysia) | **SMK Canossian Convent**
2015 – 2020
Kluang, Malaysia

HONORS, AWARDS & CERTIFICATIONS

- **InIIC 2024 Bronze Award:** Bronze Medal Award & Conference Thesis Publication — 2024 International Invention, Innovation & Creative (InIIC) Conference (Project: Patient Care Real-Time Management System).
- **NVIDIA Certification:** NVIDIA Certificate of Competency: Fundamentals of Accelerated Computing with CUDA C/C++ (Issued Sept 2023).
- **Dean's List:** Southern University College Academic Dean's List for exceptional scholastic performance across multiple trimesters.

LANGUAGES

English (Professional Working) • **Chinese / Mandarin** (Native / Bilingual) • **Malay** (Professional Working)